

Lab - Encrypting and Decrypting Data Using OpenSSL

Objectives

Part 1: Encrypting Messages with OpenSSL

Part 2: Decrypting Messages with OpenSSL

Background / Scenario

OpenSSL is an open source project that provides a robust, commercial-grade, and full-featured toolkit for the Transport Layer Security (TLS) and Secure Sockets Layer (SSL) protocols. It is also a general-purpose cryptography library. In this lab, you will use OpenSSL to encrypt and decrypt text messages.

Note: While OpenSSL is the de facto cryptography library today, the use presented in this lab is NOT recommended for robust protection. Below are two security problems with this lab:

- 1) The method described in this lab uses a weak key derivation function. The ONLY security is introduced by a very strong password.
- 2) The method described in this lab does not guarantee the integrity of the text file.

This lab should be used for instructional purposes only. The methods presented here should NOT be used to secure truly sensitive data.

Required Resources

= Security Workstation virtual machine

Instructions

Part 1: Encrypting Messages with OpenSSL

OpenSSL can be used as a standalone tool for encryption. While many encryption algorithms can be used, this lab focuses on AES. To use AES to encrypt a text file directly from the command line using OpenSSL, follow the steps below:

Step 1: Encrypting a Text File

- a. Log into Security Workstation VM.
- b. Open a terminal window.
- c. Because the text file to be encrypted is in the `/home/analyst/lab.support.files/` directory, change to that directory:

```
[analyst@secOps ~]$ cd ./lab.support.files/  
[analyst@secOps lab.support.files]$
```

- d. Type the command below to list the contents of the encrypted `letter_to_grandma.txt` text file on the screen:

```
[analyst@secOps lab.support.files]$ cat letter_to_grandma.txt
```

Hi Grandma,

I am writing this letter to thank you for the chocolate chip cookies you sent me. I got them this morning and I have already eaten half of the box! They are absolutely delicious!

```
I wish you all the best. Love,  
Your cookie-eater grandchild.  
[analyst@secOps lab.support.files]$
```

- e. From the same terminal window, issue the command below to encrypt the text file. The command will use AES-256 to encrypt the text file and save the encrypted version as **message.enc**. OpenSSL will ask for a password and for password confirmation. Provide the password as requested and be sure to remember the password.

```
[analyst@secOps lab.support.files]$ openssl aes-256-cbc -in  
letter_to_grandma.txt -out message.enc  
enter aes-256-cbc encryption password:  
Verifying - enter aes-256-cbc encryption password:  
[analyst@secOps lab.support.files]$
```

Document the password.

- f. When the process is finished, use the **cat** command again to display the contents of the **message.enc** file.

```
[analyst@secOps lab.support.files]$ cat message.enc
```

Did the contents of the **message.enc** file display correctly? **ANSWER: Yes, the file displayed correctly. What does it look like? The message was encrypted with different characters.**

- g. To make the file readable, run the OpenSSL command again, but this time add the **-a** option. The **-a** option tells OpenSSL to encode the encrypted message using a different encoding method of Base64 before storing the results in a file.

Note: Base64 is a group of similar binary-to-text encoding schemes used to represent binary data in an ASCII string format.

```
[analyst@secOps lab.support.files]$ openssl aes-256-cbc -a -in  
letter_to_grandma.txt -out message.enc  
enter aes-256-cbc encryption password:  
Verifying - enter aes-256-cbc encryption password:
```

- h. Once again, use the **cat** command to display the contents of the, now re-generated, **message.enc** file:

Note: The contents of **message.enc** will vary.

```
[analyst@secOps lab.support.files]$ cat message.enc  
U2FsdGVkX19ApWyrn8RD5zNp0RPCuMGZ98wDc26u/vmj1zyDXobGQhm/dDRZasG7  
rfnth5Q8NHValEw8vipKGM66dNFyYr9/hJUzCoqhFpRHgNn+Xs5+T0tz/QCPN1bi  
08LGTSzOpfkg76XDCK8uPy1hl/+Ng92sM5rgMzLXfEXTaYe5UgwOD42U/U6q73pj  
a1ksQrTWsv5mtN7y6mh02Wobo3AlooHrM7niOwKla3YKrSp+ZhYzVTrtksWDl6Ci  
XMufkv+FOGn+SoEEuh7l4fk0LIPEfGsExVFB4TGdTizQApRw74rTAZaE/dopaJn0  
sJmR3+3C+dmgzZiKEHwsJ2pgLvJ2Sme79J/XxwQVNpw=  
[analyst@secOps lab.support.files]$
```

Is **message.enc** displayed correctly now? The message was encoded differently.

Can you think of a benefit of having **message.enc** Base64-encoded? **Answer: Yes, It converts binary/encrypted data into plain text so it can be safely transmitted or stored in text-based systems.**

Part 2: Decrypting Messages with OpenSSL

With a similar OpenSSL command, it is possible to decrypt **message.enc**.

- a. Use the command below to decrypt message.enc:

```
[analyst@secOps lab.support.files]$ openssl aes-256-cbc -a -d -in message.enc  
-out decrypted_letter.txt
```

- b. OpenSSL will ask for the password used to encrypt the file. Enter the same password again.

- c. When OpenSSL finishes decrypting the **message.enc** file, it saves the decrypted message in a text file called **decrypted_letter.txt**. Use the **cat** display the contents of **decrypted_letter.txt**:

```
[analyst@secOps lab.support.files]$ cat decrypted_letter.txt
```

Was the letter decrypted correctly? Answer: Yes

The command used to decrypt also contains -a option. Can you explain? Answer: Yes, the -a secures the letter with password to be entered in order for the letter to be decrypted.

```

Cybersecurity LabVM Workstation 20250405 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Terminal

*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.
[analyst@secOps lab.support.files]$ cat message.enc
Salted__-----16-----qqqqqqqq[!~h74*(MeSeBee/8eeIeIeOeeXeeUeeRf ee/Ie.-H*ktCk3eH!e0!HX0e+++++e2ueAeeCpe2egm
eeefp3p3ee-----eeef eNeeEebel[eHkHnnneee e*CbIeYeDeee]JNee9e3e@eadee+ eeefJeI
Nyee*4
[analyst@secOps lab.support.files]$ openssl aes-256-cbc -in letter_to_grandma.txt -out message.enc
-a
enter AES-256-CBC encryption password:
Verflying - enter AES-256-CBC encryption password:
*** WARNING : deprecated key derivation used.
Using -iter or -pbkdf2 would be better.
a: command not found
[analyst@secOps lab.support.files]$ cat message.enc
e,e,ee50eeEve[e]bbee?PeGdneeeekkeeePe ee2MH4eeK e2)RdeeeQeehee[CeekepeIeDeeeUee?9Meeeo* ePe TecG8ek ee3_eer[e]eTe[-eezeeeYeeT[e(eee83eeeee(Aeeecv-e)ee8Dn?eHeex8ee-eeDee_e[analyst@secOps lab.support.files]$
openssl aes-256-cbc -a -d -in message.enc -out decrypted_letter.txt
[analyst@secOps lab.support.files]$ openssl aes-256-cbc -a -d -in message.enc -out decrypted_letter.txt
enter AES-256-CBC decryption password:
error reading input file
[analyst@secOps lab.support.files]$ cisco
command 'cisco' not found, did you mean:
  command 'cvsco' from deb cvsutils (0.2.5-1.1)
  command 'disco' from deb discodos (1.0-rc2-2ubuntu0.22.04.1)
  command 'disco' from deb mono-devel (6.8.0.105+dfsg-3.2)
  command 'cisco' from deb cisco (1.0.0-0ubuntu5)
  command 'cisco' from deb graphviz (2.42.2-0ubuntu0.1)
Try: sudo apt install cdeb names
[analyst@secOps lab.support.files]$ cat decrypted_letter.txt
[analyst@secOps lab.support.files]$ openssl aes-256-cbc -a -d -in message.enc -out decrypted_letter.txt
enter AES-256-CBC decryption password:
error reading input file
[analyst@secOps lab.support.files]$ cat decrypted_letter.txt
[analyst@secOps lab.support.files]$ openssl aes-256-cbc -a -d -in message.enc -out decrypted_letter.txt
-a
enter AES-256-CBC decryption password:
error reading input file
a: command not found
[analyst@secOps lab.support.files]$ openssl aes-256-cbc -a -d -in message.enc -out decrypted_letter.txt
-a
enter AES-256-CBC decryption password:
error reading input file
a: command not found
[analyst@secOps lab.support.files]$

```